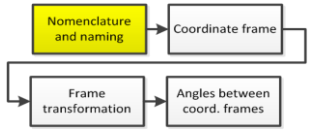


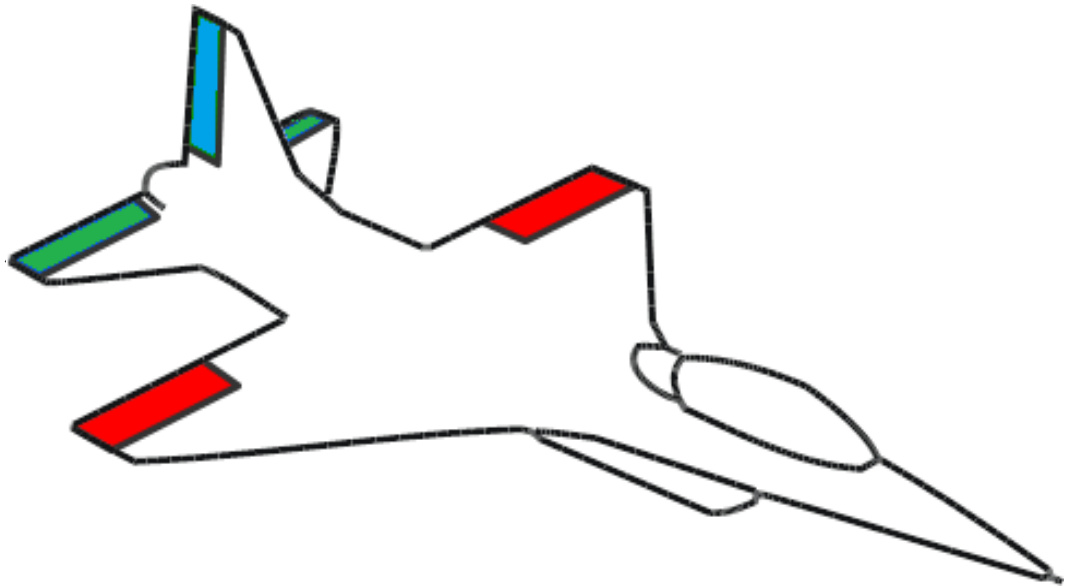
Introduction to Flight Mechanics & Control

Exercise Session 1

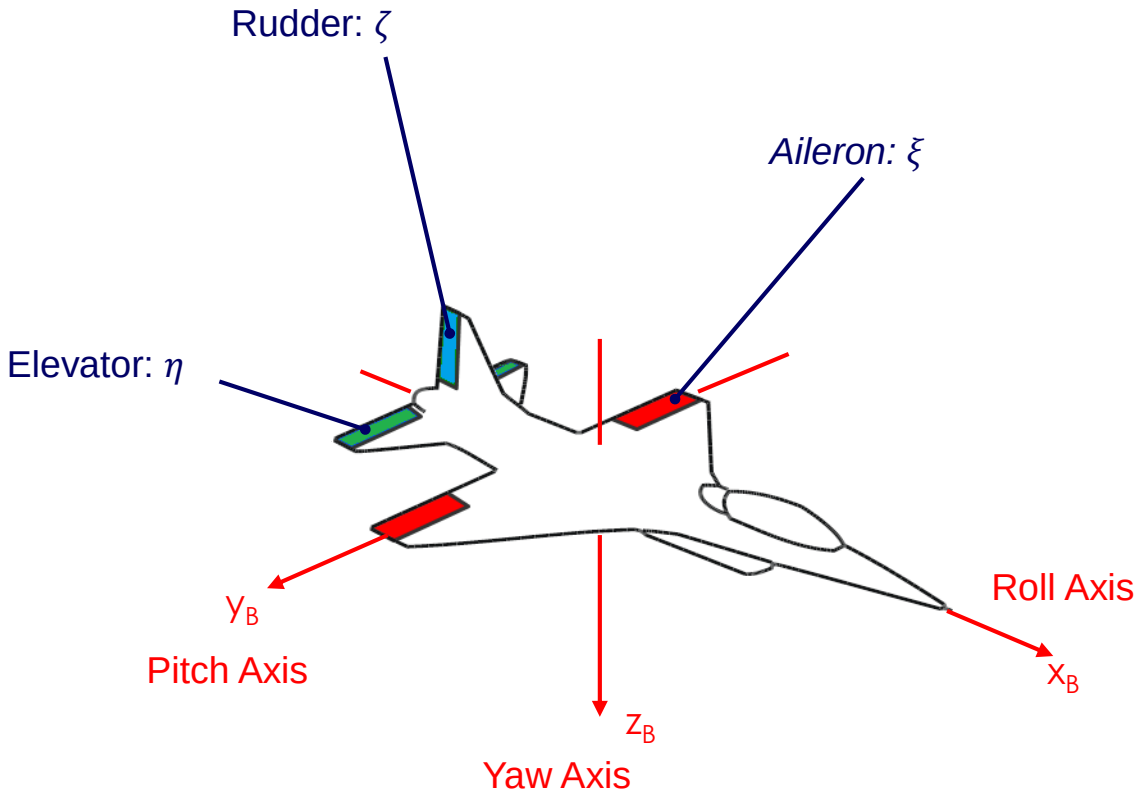
Aerodynamic Surfaces



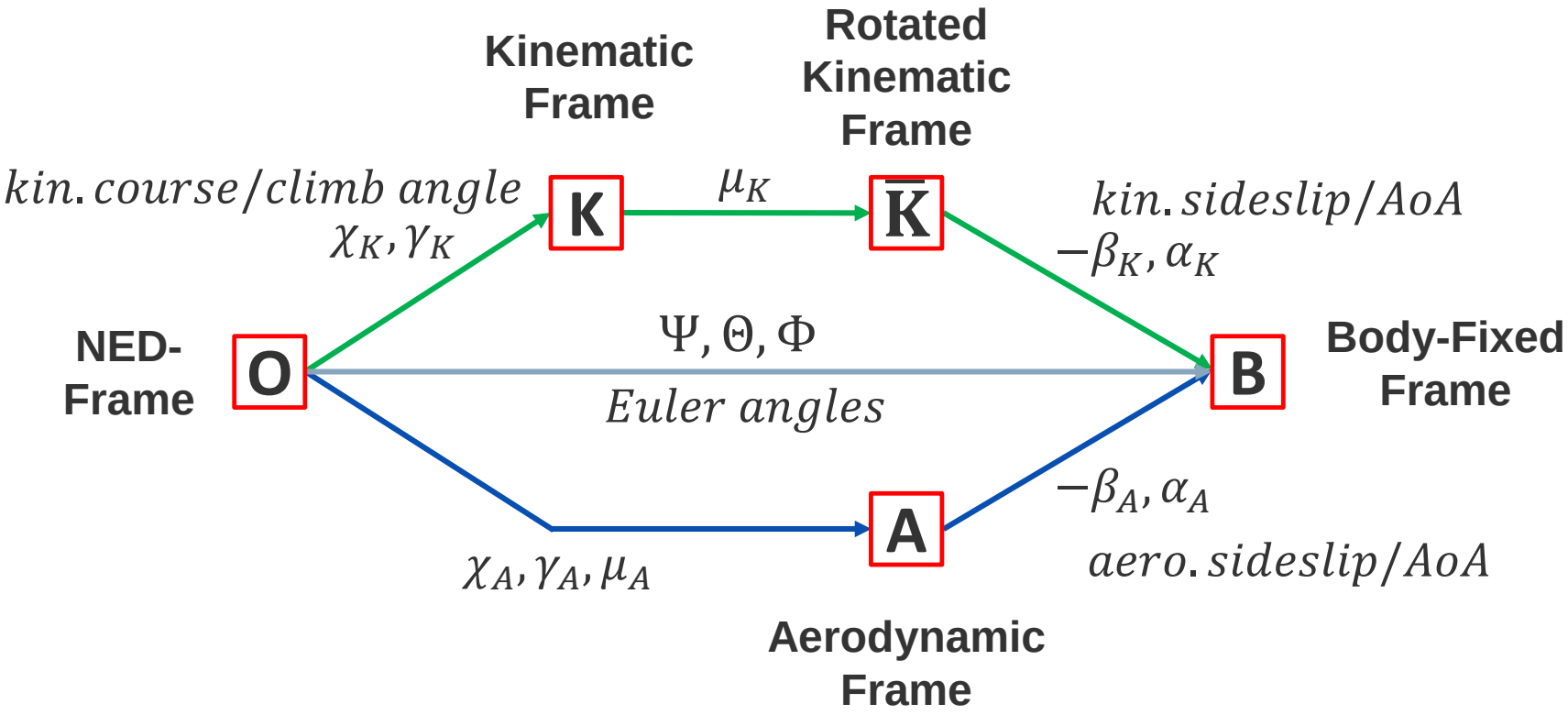
Give the symbol and name of the **primary aerodynamic control surfaces**. Give the symbol and the name of the **body fixed axis about which the main moment is produced by each of the surfaces**.



Aerodynamic Surfaces cont'd



B Body-Fixed Frame


Kinematic Frame
Rotated Kinematic Frame
kin. course/climb angle
χ_K, γ_K
K
μ_K
$\bar{K}$
kin. sideslip/AoA
$-\beta_K, \alpha_K$
NED-Frame
O
Ψ, Θ, Φ
Euler angles
B
Body-Fixed Frame
$-\beta_A, \alpha_A$
A
χ_A, γ_A, μ_A
aero. sideslip/AoA
Aerodynamic Frame

The position of a point in ECEF-frame is either given in cartesian coordinates x, y, z or in WSG-84 coordinates, which are composed of two angles and the height over the reference ellipsoid

WGS-84 coordinates are relative to ECEF-frame.

Geodetic Longitude: λ

angle, measured in Equatorial plane, between prime meridian (xz-plane in ECEF-frame) and the meridial plane of point *P*.

Range: $-\pi \leq \lambda < \pi$

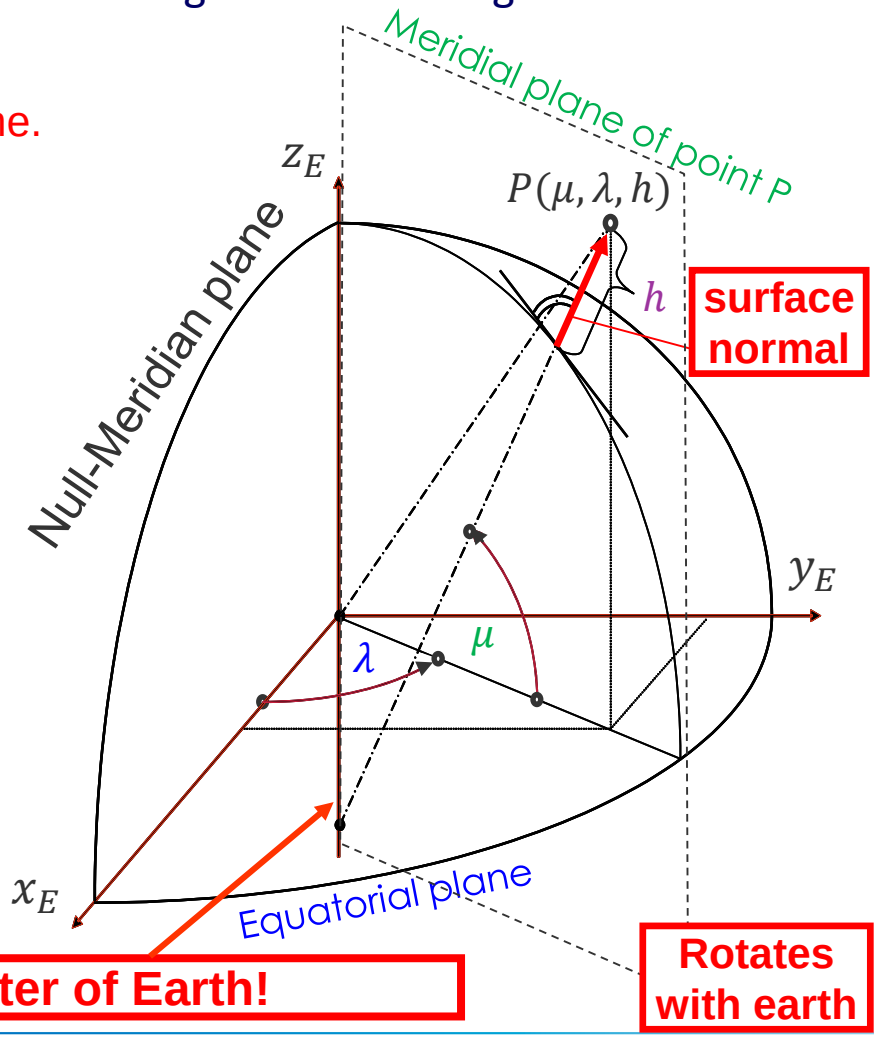
Geodetic Latitude: μ

angle, measured on meridial plane of point P, between Equatorial plane (xy-plane of ECEF-frame) and the surface normal point P.

Range: $-\pi/2 \leq \mu \leq \pi/2$

Geodetic Height: h

Height over WGS-84 ellipsoid along surface normal.



Surface normal does not pass the center of Earth!

Transformation of WGS-84 coordinates to cartesian ECEF coordinates:

$$(\vec{r})_{E(WGS\,84)} = \begin{bmatrix} \lambda \\ \mu \\ h \end{bmatrix} \Longrightarrow (\vec{r})_E = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$x = (N + h) \cdot \cos\mu \cdot \cos\lambda$$
$$y = (N + h) \cdot \cos\mu \cdot \sin\lambda$$
$$z = \left[N\left(1 - e^2\right) + h\right] \cdot \sin\mu$$

Length of semi-major axis: $a = 6378137{,}0\ m$

Length of semi-minor axis: $b = 6356752{,}3142\ m$

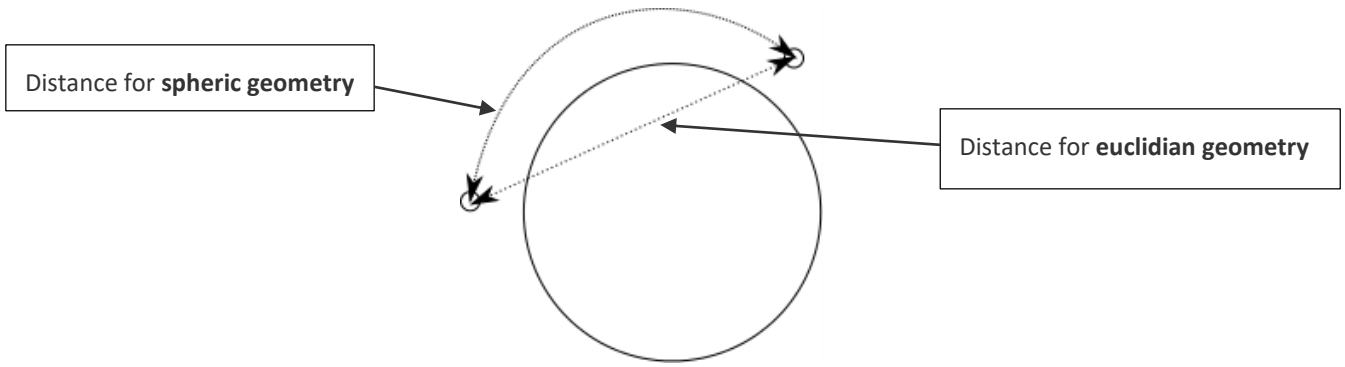
Earth flattening: $f = \frac{a-b}{a} = 0{,}0034$

First excentricity: $e = \sqrt{f(2-f)} = 0{,}0818$

Prime vertical radius: $N(\mu) = \frac{a}{\sqrt{1-e^2\,\sin(\mu)^2}}$

Note:
The transformation of ECEF coordinates to WGS-84 coordinates has to be done iteratively!

Exercise 1.2
Comparison spheric vs. euclidian geometry



Distance of two points on the earth surface with spheric geometry (Assumption of mean earth radius)	Distance of two points on the earth surface with euclidian geometry (Assumption of mean earth radius)	Difference
87,4133120893057 m	87,4132743842732 m	-3,77050324260608e-05 m
1008,60102766933 m	1008,60102414953 m	-3,51980042978539e-06 m
4920,22163961305 m	4920,22151833256 m	-0,000121280487292097 m
9783,66070511574 m	9783,65974560852 m	-0,000959507220613887 m
113903,199169905 m	113901,685583886 m	-1,51358601891843 m
1204418,23726070 m	1202629,52922606 m	-1788,70803463738 m

Error α Distance

Key Facts:

1. Aircraft with constant velocity
2. Goal: Find kinematic course angle χ_K and kinematic flight path angle γ_K
3. Given:

   a. GPS-Position of the aircraft at two points in time:

t_1 = 0 s:

Geodetic Longitude	λ_1	11.669284 deg
Geodetic Latitude	μ_1	48.265906 deg
Geodetic/GPS Height	h_1	1500 m

t_2 = 60 s:

Geodetic Longitude	λ_2	11.591421 deg
Geodetic Latitude	μ_2	48.22776 deg
Geodetic/GPS Height	h_2	1530 m

   b. Length of semi major axis: $a_{earth} = 6378137$ m

   c. First excentricity : $e_{earth} = 0.0818$


Point_t_1
Point_t_2
Ruler
Line
Path
Polygon
Circle
3D path
3D polygon
Measure the distance between two points on the ground
Map Length:
7,171.52
Meters
Ground Length:
7,172.08
Heading:
233.69 degrees
Mouse Navigation
Save
Clear

a) Calculate the position vectors of both points in the ECEF system.

$$\left(\vec{r}_1^{\,G}\right)_E = \begin{bmatrix} x_1^G \\ y_1^G \\ z_1^G \end{bmatrix}_E = \begin{bmatrix} (N_1 + h_1)\cos\mu_1 \cos\lambda_1 \\ (N_1 + h_1)\cos\mu_1 \sin\lambda_1 \\ \left(N_1\left(1-e^2\right) + h_1\right)\sin\mu_1 \end{bmatrix}_E = \begin{bmatrix} 4.16685e\ 6 \\ 8.6056e\ 5 \\ 4.7377e\ 6 \end{bmatrix}_E$$

$$\left(\vec{r}_2^{\,G}\right)_E = \begin{bmatrix} x_1^G \\ y_1^G \\ z_1^G \end{bmatrix}_E = \begin{bmatrix} (N_2 + h_2)\cos\mu_2 \cos\lambda_2 \\ (N_2 + h_2)\cos\mu_2 \sin\lambda_2 \\ \left(N_2\left(1-e^2\right) + h_2\right)\sin\mu_2 \end{bmatrix}_E = \begin{bmatrix} 4.1710e\ 6 \\ 8.5554e\ 5 \\ 4.7349e\ 6 \end{bmatrix}_E$$

$$N(\mu) = \frac{a}{\sqrt{1 - e^2\,\sin(\mu)^2}}$$

$$N_2 = \frac{6378137}{\sqrt{1 - 0.0818^2\ \mathrm{si\,n}(\,0.8417)^2}} = 6390039.226359209$$

$$N_1 = \frac{6378137}{\sqrt{1 - 0.0818^2\ \mathrm{si\,n}(\,0.8424)^2}} = 6390053.421249251$$

Calculate the velocity vector of the aircraft by using both position vectors from a). Use a suitable and intuitive coordinate system.

$$\left(\vec{V}_x^G\right)_E^E = \frac{1}{\Delta t}\left(\left(\vec{r}_2^G\right)_E - \left(\vec{r}_1^G\right)_E\right) = \frac{1}{60}\left(\begin{bmatrix}4.1710e\ 6\\ 8.5554e\ 5\\ 4.7349e\ 6\end{bmatrix}_E - \begin{bmatrix}4.16685e\ 6\\ 8.6056e\ 5\\ 4.7377e\ 6\end{bmatrix}_E\right) = \begin{bmatrix}71.4293\\ -83.7215\\ -46.7151\end{bmatrix}_E$$

Why do we not use the E frame?

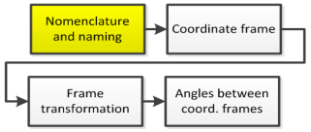
$$\left(\vec{V}_x^G\right)_O^E = \boldsymbol{M}_{OE} \cdot \left(\vec{V}_x^G\right)_E^E = \begin{pmatrix} -\sin\mu\cos\lambda & -\sin\mu\sin\lambda & \cos\mu \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\mu\cos\lambda & -\cos\mu\sin\lambda & -\sin\mu \end{pmatrix} \cdot \begin{bmatrix}71.4293\\ -83.7215\\ -46.7151\end{bmatrix}_E$$

$$= \begin{bmatrix}-70.6623\\ -96.4385\\ -0.4328\end{bmatrix}_O$$

Here, the points are not far off each other.
You can therefore use either one of the points in the trafo matrix….

$$\left|\left(\vec{V}_x^G\right)_O^E\right| = 119.556\ \text{m/s}$$

What type of velocity is the value calculated in b)?.



- **Velocity of point *G* relative to the center of Earth *O***

$$\left(\frac{d}{dt}\right)^{E}\left(\vec{r}^{G}\right)_{B}=\left(\vec{V}_{K}^{G}\right)_{B}^{E}=\begin{bmatrix}u_K^G\\ v_K^G\\ w_K^G\end{bmatrix}_{B}^{E}$$

Velocity relative to the *ECEF-frame*

Components in body fixed frame

- **Velocity of point *P* relative to point *G***

$$\left(\frac{d}{dt}\right)^{I}\left(\vec{r}^{GP}\right)_{B}=\left(\vec{V}_{K}^{GP}\right)_{B}^{I}=\begin{bmatrix}u_K^{GP}\\ v_K^{GP}\\ w_K^{GP}\end{bmatrix}_{B}^{I}$$

Velocity relative to the *ECI-frame*

Components in body fixed frame

Types of velocity

***K* – kinematic** **_W_ – wind** ***A* – aerodynamic**

Velocities are always given in m/s

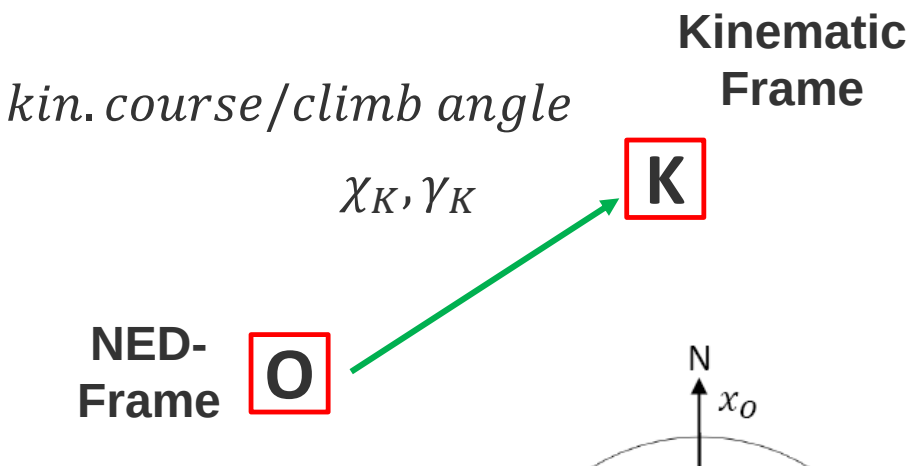
What type of velocity is the value calculated in b)?.

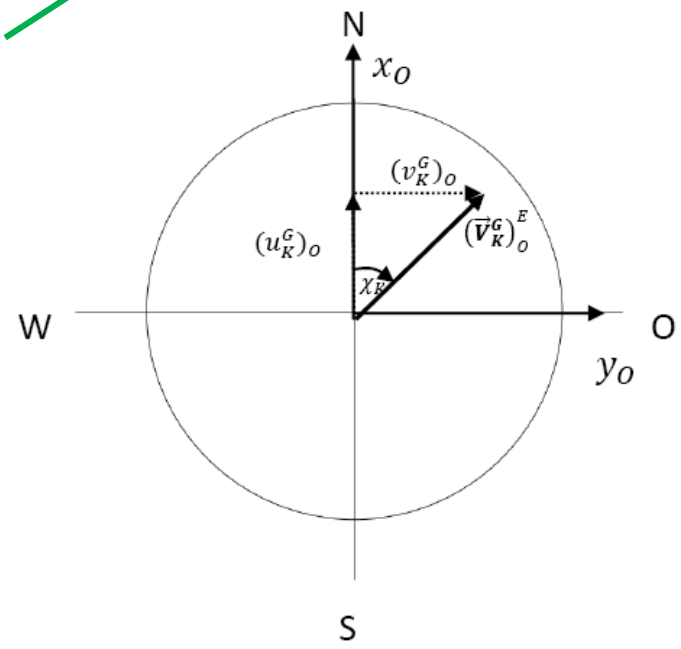
Recall:

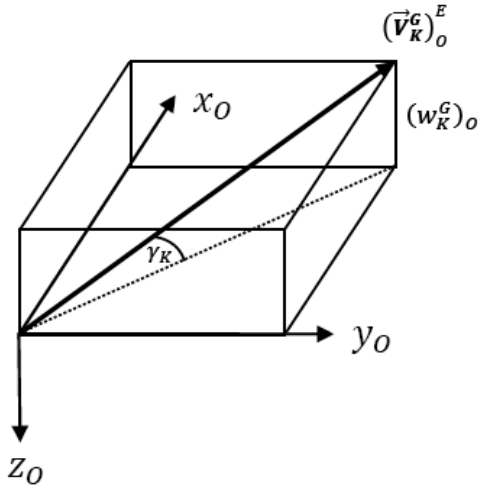
- we computed the velocity from the change of position in the ECEF frame
- we state the velocity in the NED (O) frame
- as we have no information about the wind condition, this must be the kinematic speed

$$\left(\vec{V}_{K}^{G}\right)_{O}^{E}$$

d) Calculate the angles χ_K and γ_K in point 1 (point for time t_1).







d) Calculate the angles χ_K and γ_K in point 1 (point for time t_1).

$$\gamma_K = \operatorname{asin}\frac{-\left(w_K^G\right)_O}{\left|\left(\vec{V}_K^G\right)_O^E\right|} = \operatorname{asin}\frac{0.4328}{119.556} = 0.0036\ \text{rad} = 0.2074\ °$$

$$\chi_K = \operatorname{atan}\frac{\left(v_K^G\right)_O}{\left(u_K^G\right)_O} = \operatorname{atan}\frac{-96.4385}{-70.6623} = 0.9384\ \text{rad} = 53.7691\ °$$

Problem: atan only defined within -90 to 90 degrees, we are interested in a 360 degree course range!

Solution: use atan2 (Matlab function) which leads to

$$\chi_K = 53.7691\ ° + 180° = 233.7691°$$

e) Give an easy way to check the results of exercise d).

$$\left(\vec{V}_K^G\right)_K^E = M_{KO} \cdot \left(\vec{V}_K^G\right)_O^E = \begin{pmatrix} \cos\chi_K \cdot \cos\gamma_K & \sin\chi_K \cdot \cos\gamma_K & -\sin\gamma_K \\ -\sin\chi_K & \cos\chi_K & 0 \\ \cos\chi_K \cdot \sin\gamma_K & \sin\chi_K \cdot \sin\gamma_K & \cos\gamma_K \end{pmatrix} \begin{bmatrix} -70.6623 \\ -96.4385 \\ -0.4328 \end{bmatrix}_O$$

$$\left(\vec{V}_K^G\right)_K^E = \begin{bmatrix} 119.5564 \\ 0 \\ 0 \end{bmatrix}_K$$